Silent Reasoning at 124M: A Pre-registered, Cost-Accounted Test of Continuous Latent Reasoning Against Tuned Chain of Thought

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Abstract

Continuous latent reasoning (Coconut) feeds a decoder its own last hidden state as the next input embedding instead of a written token. The published claim is that this matches or beats written chain of thought (CoT) with far fewer emitted tokens. We test that claim at GPT-2 124M under a pre-registered judge with a cost-accounted, seeded design, an independent automated reviewer, and kill rules written before any run. First, the reproduction passes: we match the published Coconut, CoT, No-CoT and pause cells on ProsQA (three seeds for CoT and Coconut) and the CoT and Coconut cells on GSM8k-Aug (one seed) within the registered tolerances. Second, on two harder screens at one seed (ProsQA-hard and a 50k GSM8k-Aug subset) every curriculum-trained latent arm, including three training objectives we add, ties or trails a pause-token control trained on the same curriculum and sits 7.0 to 16.2 points under a 3-learning-rate tuned CoT bar; no arm advances. Third, the reproduced Coconut checkpoints' latents are idle under intervention on ProsQA (scrambled drop 0.0 on every seed) and load-bearing on GSM8k-Aug (scrambled drop 30.71 points). Fourth, a latent-dependence hinge loss makes the model break when its own latents are replaced (scrambled drop +10.4 and +11.8 points) but the model never reads another example's content: the donor's answer appears 0 of 289 times. The hinge trains a self-signature, not a readable state. Fifth, at one seed, running latent steps through 8 of 12 layers keeps full-depth accuracy within the screens' tie band (3.60 points, a ProsQA CoT seed SD) on both tasks at a measured per-step wall-clock fraction of 0.7200 and 0.7280. Every latent arm meets the FLOP and token legs and fails the sequential-passes criterion, so every token saving is a billing claim only. The result is a clean negative within its scope (124M, seed 0, validation-split screens).

1 Introduction

Hao et al. (2024) showed that a GPT-2 124M model can reason in continuous latent steps. At each latent position the input embedding is the previous position's last-layer state, fed back without a vocabulary bottleneck. On ProsQA the published Coconut scores 97.0 against CoT's 77.5 with 14.2 emitted tokens against 49.4; on GSM8k-Aug it scores 34.1 against CoT's 42.9 with 8.2 tokens against 25.0. Since then, independent groups have shown that the ProsQA gain travels with the training curriculum (a pause-token control under the same curriculum reaches 96.6), that the latent vectors of fine-tuned Coconut change few answers when perturbed, and that single-vector latents collapse past a handful of steps (Section 2).

The claim we test is a strong form of the efficiency promise: match or beat a *tuned* written-CoT model on verifiable tasks with at least 5× fewer emitted tokens at equal total FLOPs, with latents that carry state under intervention, and without losing to a bigger plain model at equal FLOPs. Two facts shape the test. A latent step costs a forward pass of the whole model at one position, the same as a written token minus the LM head, so "fewer tokens at equal compute" is a billing claim unless steps are fewer or cheaper. And the recent literature says the flagship win is the curriculum, so any new method must *force* its latents to carry state, and the readout must be judged by intervention.

We see five open gaps in the literature. (1) We found no seeded comparison of latent reasoning against a tuned CoT of the same base that charges every arm its FLOPs, sequential passes and emitted tokens; published comparisons do not match compute between arms. (2) Theory says a single-vector latent is a bounded channel; we found no test of the width hypothesis at matched FLOPs. (3) Latents are idle under intervention in fine-tuned Coconut, and we know of no objective for continuous latents that trains directly against that failure using the scrambled-latent control itself (a counterfactual-rationale loss exists for written CoT; Section 2). (4) Scaling evidence is thin and mixed: prior latent

methods trail explicit CoT past 1B with a widening gap, and one looped method closes it at 3B (Fan et al., 2026). (5) We found no latent-reasoning result on program-output prediction. This paper addresses (1) and (3) at 124M; (2), (4) and (5) are untested here.

Our contributions:

1. A pre-registered judge for latent reasoning: a tuned CoT bar with trial parity, pause-same-curriculum and scrambled-at-inference nulls, a plain-depth control at equal FLOPs, three cost axes charged per example (FLOPs, sequential passes, emitted tokens), kill rules K1–K3, and an independent automated reviewer that wrote no model code and recomputed every verdict from hashed per-example files.
2. A reproduction of Coconut on GPT-2 124M that matches the published cells within tolerance, with a per-seed intervention diagnostic: latents idle on ProsQA, load-bearing on GSM8k-Aug. The reproduction itself is not a contribution; the seed spread and the diagnostic are.
3. Screens of three training objectives on top of Coconut: partial-depth latent steps (C1b), verifiable-state supervision (C2), a latent-dependence hinge (C3), and C2+C3, against seven controls on two tasks, with a no-curriculum control on both tasks. None advances; every latent arm ties the pause control or trails it.
4. A directional read of the hinge's scrambled drop: the model detects foreign latents and does not read their content (donor answer 0 of 289), which we call a self-signature.
5. A measured per-step cost saving: latent steps through the middle 8 of 12 layers (layer skipping is itself known; Fan et al., 2020; Elhoushi et al., 2024) keep full-depth accuracy within σ at a measured wall-clock fraction of 0.7200 (GSM-50k) and 0.7280 (ProsQA-hard) of a full-depth latent pass at batch 1.
6. The complete token, FLOP and pass accounting of every arm, showing that the 13× token saving of the latent arms on GSM-50k comes with a sequential-pass count above the registered criterion on every feedback arm.

The answer "silent reasoning does not beat written reasoning at this size and budget" was listed in advance as an acceptable outcome. Our prior, written before any run, put a clean negative at about 0.60.

2 Related work

Numbers in this section were read from the cited papers or their official code.

Coconut. Hao et al. (2024; COLM 2025) train GPT-2 124M with a staged curriculum: stage k removes the first k reasoning steps and inserts $k \times c_{\text{thought}}$ latent positions. Their Table 1 gives ProsQA Coconut 97.0 ± 0.3, pause-as-thought 96.6 ± 0.8, CoT 77.5 ± 1.9, No-CoT 76.7 ± 1.0, Coconut without curriculum 76.1 ± 0.2; GSM8k-Aug Coconut 34.1 ± 1.5, CoT 42.9 ± 0.2, No-CoT 16.5 ± 0.5, pause-as-thought 24.1 ± 0.7.

The 2026 critiques. An unreviewed preprint produced with an automated research pipeline (Anonymous 2026; checkpoints at `https://huggingface.co/bmarti44/coconut-curriculum-checkpoints`) retrained Coconut and a pause-token control under the same seven-stage curriculum: 96.6 against 97.0, McNemar $p = 0.845$. Aswal et al. (2026) retrained Coconut and CODI on ProsQA with controls lacking recurrence or curriculum: BFS-like frontiers and decodable arithmetic appear in the controls too, and the causal use of a latent is graded and low-rank. Rizvi-Martel et al. (COLM 2026) find that fine-tuned Coconut takes the answer from the question representation; superposition appears only when trained from scratch. Zhang et al. (2025) find perturbing Coconut latents flips under 5–10% of answers against up to 50% for CoT tokens. Dilgren and Wiegreffe (2026) find latents removable on logic tasks and more important on math. Jin et al. (2026) find counterfactual responsiveness falling from 71.3 to 27.4 along training while accuracy rises from 73.1 to 90.1. Backour (2026) shows softmax readout plus autoregressive feedback collapses or amplifies mixtures. Li et al. (ICML 2026) find latent steps act as staged functionality with non-local routing. These studies are why our judge requires the pause-same-curriculum control, the scrambled-latent control, zero-latent and step sweeps, and an intervention-based readout.

Methods that improved on Coconut. CODI (Shen et al., EMNLP 2025) distils the explicit CoT into the student's hidden state in one stage: GPT-2 GSM8k-Aug about 43.7 against CoT-SFT 44.1 at 3.1× compression. SIM-CoT (Wei et al., ICLR 2026) adds an auxiliary decoder per latent step: +8.2 over Coconut on GPT-2; it reports that Coconut training becomes unstable and sometimes collapses as the implicit tokens grow from the default three to five. KaVa (Kuzina et al., ICLR 2026) distils a compressed KV cache and stays below CoT at every size. LOTUS (Fan et al., 2026) loops a padded transformer with per-position supervision and states that prior latent methods trail explicit CoT past 1B with a widening gap; it closes the gap at 3B (70.0 against 71.5). PCCoT (Wu et al., EMNLP 2025) updates latent slots in parallel by Jacobi iteration. MUX (Suleymanzade et al., 2026) multiplexes subword spans into one latent. CoLaR (Tan

et al., NeurIPS 2025) compresses chains with a promptable factor. The common thread: the methods that close the gap supervise each latent step against the explicit step it replaces, or widen the per-step state.

Making latents causal. Ray et al. (2025) train latent "thinking" tokens with an outcome reward in a final reinforcement-learning stage, arguing that answer-only supervision lets a model bypass its latents. Zhang et al. (2025) analyse Coconut's latents causally and adversarially and conclude that they mostly exploit shortcuts. For written CoT, FRODO (Paul et al., 2024) trains the answer model with a margin-ranking loss that makes the answer more likely under a correct rationale than under a counterfactual one; this is the closest antecedent of our hinge. Our hinge (C3) applies that kind of margin to fed-back continuous latents, with another example's latents as the counterfactual: it uses the scrambled-latent control itself as a training signal, rewarding the model when its own latents make the answer more likely than another example's latents do.

Recurrent depth and pause tokens. Saunshi et al. (ICLR 2025) show k layers looped L times nearly match a kL-layer model; Huginn (Geiping et al., 2025) scales a prelude-core-coda model to 3.5B; Ouro (Zhu et al., 2025b) pretrains looped 1.4B and 2.6B models. Goyal et al. (ICLR 2024) find pause tokens help only when used in pretraining; fine-tune-only gains are mild. In a separate, unpublished experiment on BabyLM data we found a 12-layer model looped twice lost 0.038 nats to a plain 24-layer model at equal FLOPs, which is why a plain-depth control is mandatory here.

Formal results. Zhu et al. (NeurIPS 2025a) prove two layers with D continuous thoughts solve reachability of diameter D where constant-depth discrete CoT needs $O(n^2)$ steps. Xu and Sato (ICML 2026) show latent thought parallelises what CoT serialises, while CoT wins on counting and sampling. Zhang (2026) shows that, under a standard complexity assumption, a fixed-width recurrent state cannot decide P-complete problems under logspace reductions, while polynomial CoT can. Chen et al. (ICML 2026) show outcome-only supervision gives dual collapse and reconstruction objectives preserve $I(\text{latent}; \text{step})$. Zou et al. (2026) argue the curriculum is necessary.

What this paper adds. A pre-registered, cost-accounted test against a *tuned* CoT of the same base, with all the controls above on one judge; a training objective that attacks latent idleness directly; a directional read that separates detection from reading; and a measured per-step cost fraction for partial-depth latent steps.

3 Setup

3.1 Tasks

- **ProsQA** (released with Coconut): 17,886 / 300 / 500 records; graph reachability with the edges in the question; about 180 words; 3 proof steps. **GSM8k-Aug** (the augmented set of Deng et al. 2023, in the iCoT release of Deng et al. 2024): 385k train; valid 500; test = GSM8K test, 1,319.
- **ProsQA-hard** (ours): proof depth drawn uniformly from {5, 6, 7}, 1.7× the released edge count, about 300-word questions; 10,000 / 300 / 500 from one seeded stream; no duplicate question across splits. 13-gram overlap with its own train is 37.0% on valid (template sentences over a tiny vocabulary; not leakage; Data: `data/table9_text_numbers.csv`).
- **GSM-50k** (ours): the first 50,000 records of a seed-0 permutation of the GSM8k-Aug train set; valid the full 500; test the full 1,319. The valid set shares a 13-gram with the full GSM8k-Aug train set on 7 of 500 items, equal for every arm.

All tasks are scored by exact match with the reference implementation's answer extraction (`split("#")[-1]`, commas stripped), greedy decoding, `max_new_tokens` 64 (GSM) or 128 (ProsQA).

3.2 Model and recipes

Base models: `openai-community/gpt2` (124M) and `gpt2-medium` (355M), both pinned to a fixed hub revision and hash-verified at every launch. Phase 1 (the reproduction) follows the published recipe: AdamW lr 1e-4, weight decay 0.01, effective batch 128, fp32 with TF32 matmuls (fp32-tf32; a precision pair shows $d_{\text{prec}} = 0.0$ against strict fp32); ProsQA c_{thought} 1, 5 epochs per stage, 6 latent stages, 50 epochs; GSM8k-Aug a 25-epoch CoT stage, then Coconut from that checkpoint with c_{thought} 2, 3 stages of 3 epochs. Phase 2 (the screens) runs bf16 autocast over TF32 on A100-80GB spot VMs. ProsQA-hard latent arms: c_{thought} 1, 3 epochs per stage, 7 stages, 30 epochs, selection over epochs 22 to 30, read with 7 latents. GSM-50k latent arms: start from the lr 1e-4 CoT run's final checkpoint, c_{thought} 2, 3 stages of 3 epochs, 25 epochs, selection over epochs 13 to 25, read with 6 latents. Selection is on valid only; the test split is read once at the end of the round for reporting.

Table 1: Arms, cards and their nulls. A card is a new method registered with its own pass line (C1b, C2, C3 and C2+C3); every other arm is a control.

arm	what it is	its null or control	registered question
`cot` (bar)	written CoT, 3 LR trials {3e-5, 1e-4, 3e-4}, best on valid		the bar for every comparison
`no_cot`	answer only		the floor; the 124M plain-depth point
`coconut`	full-depth Coconut, the published recipe	pause, scrambled, zero-latent, step sweep	the reference latent arm
`pause`	pause tokens under the same curriculum and token budget		curriculum vs content
scrambled-at-inference	the arm's checkpoint with every latent replaced by a donor example's latents (seeded derangement, bit for bit)		K2: does the content matter
zero-latent; step sweep {0, 1, 2, 4, 8}	the checkpoint read with no or n latents		dose-response; decides nothing
plain depth	GPT-2 355M `no_cot`, same data and epochs		K3: a bigger plain model at equal FLOPs
C1b $k = 8$, $k = 4$	latent steps through the band of blocks [2, 10) or [4, 8) of 12; non-band blocks skipped, their KV masked	full-depth `coconut`	a cheaper step with accuracy held
C2	Coconut + an auxiliary 2-layer MLP decoder on each fed-back latent, trained to reconstruct the executed intermediate state (ProsQA frontier; GSM step result), λ 0.1, dropped at inference	C2-shuffled: the same decoder on a different example's state	does correct state content help
C3	Coconut + a second pass with a scrambled bank; loss $\mathrm{CE}_{\mathrm{true}} + \max(0,\ 0.2 - (\mathrm{CE}_{\mathrm{scr}} - \mathrm{CE}_{\mathrm{true}}))$ per answer token	C3-foreign: the same hinge between two foreign banks A and B	can dependence on own latents be trained in
C2+C3	both objectives on one model	attribution set: C2, C3, `coconut`	the arm most likely to pass

Data: see `data/`.

3.3 Arms and their nulls

Every arm stores its latents as positions in the KV cache (an appended scratchpad), so the fixed-width separation results of Section 2 do not bind them. C3-foreign replaced the first proposed null for C3, a hinge at a question position, which the reviewer showed has zero gradient by causality.

3.4 The judge and the kill rules

Primary claim test. One task, one size, one latent arm against the bar, all three legs: (1) accuracy beats or ties the bar; (2) mean per-example inference FLOPs ≤ the bar's; (3) mean emitted tokens ≤ 0.2× the bar's. A secondary criterion is printed beside leg 3 at all times: sequential passes ≤ 0.2× the bar's. An arm that meets leg 3 but not the passes criterion carries the label **"billing claim only"**.

Kill rules. K1: a published cell is matched iff our value lies within max(2.0 points, 2 σ_{cell}) of it, with σ_{cell} = max(1.0, the pooled SD of our seeds ($n - 1$ degrees of freedom) and the published spread s_{pub} (2 degrees of freedom)), as the K1 code computes it; with our three seeds and SD s_{ours} this is $\max\left(1.0,\ \sqrt{(s_{\mathrm{ours}}^2 + s_{\mathrm{pub}}^2)/2}\right)$, and for a single-seed cell $\max(1.0, s_{\mathrm{pub}})$; a miss starts a debug round. K2: an arm whose own-minus-scrambled drop Δ_{scr} is under σ, or whose paired example-bootstrap interval includes 0, is dead as a latent claim and continues only as a labelled curriculum or token-budget effect; an arm within σ of pause is **CURRICULUM-ONLY**. K3: an arm below the plain-depth curve $A_{\mathrm{pd}}(F)$ by more than σ at its own FLOPs is dead as an efficiency claim; within σ is "no better than plain depth". A_{pd} is linear in $\ln F$ between the 124M and 355M `no_cot` points, with no extrapolation.

Screen rule. At seed 0 an arm advances iff it is within 1 σ_{screen} of the bar, ahead of its scrambled null and of pause by 1 σ_{screen}, and meets the FLOP and token legs. A screen filters; it is never a verdict. Here σ_{screen} = max(1.0, the binomial SE at the bar's valid accuracy, the ProsQA `cot` 3-seed SD) = 3.60 on both tasks (terms: floor 1.0; binomial

1.9517 on GSM-50k, 2.2948 on ProsQA-hard; the 3-seed SD 3.6019). So σ_{screen} is the seed SD of the reproduced CoT model on released ProsQA (three seeds), used as the tie band on both screen tasks; it is wider than the binomial SE of either screen. No latent arm has a measured seed SD on either screen task.

Hinge arms. For an arm trained against scrambled latents, K2 is not independent: the hinge maximises exactly the quantity K2 measures, and a model can satisfy it with a fingerprint (one latent checks the question; the head emits garbage on mismatch) and no reasoning state. So for these arms the K2 cell prints "K2 not independent"; the C3 minus C3-foreign drop is labelled "dependence on own latents (content or fingerprint)"; and a non-circular directional read is required. On GSM-50k that read is the **donor-answer read**: among scored examples whose donor's last supervised step result equals the donor's answer, the rate at which the generated answer under the donor's latents equals the donor's answer, against the same rate under own latents. A large drop whose directional read has an interval including 0, or whose garbage fraction exceeds its directional fraction, is "drop without direction (fingerprint-consistent)", and K2 counts as fired.

C1b wall-clock. The per-step fraction f_{meas} is the C1b checkpoint's latent milliseconds per pass over the full-depth `coconut` checkpoint's, run through the same band code at band [0, 12], batch 1, same GPU and torch build. The saving "shows" iff $f_{\text{meas}} \le (b-a)/L + 0.10$; otherwise it is "nominal" and K3 and the primary test use the measured fraction. The registration printed the lines to two decimals (0.77 and 0.43) while the formula gives 0.7667 and 0.4333; each line is read at the stricter of the two (0.7667 for $k = 8$, 0.43 for $k = 4$). One reading depends on this choice: GSM-50k C1b $k = 4$'s 0.4326 lies between 0.43 and 0.4333, so it is "nominal" at the printed line and would show at the exact one.

3.5 Cost accounting

Per example the judge charges every forward pass: prefill plus every sequential pass, latent steps and emitted tokens alike, with the LM head charged only where logits are produced (never at a latent position). With N_{ne} non-embedding parameters, $V = 50{,}260$ (after the resize; charged as each arm's own head), $d = 768$, $n_L = 12$, $d_a = 768$:

$$\texttt{flops_prefill} = 2N_{\text{ne}}\,p + 2Vd\,\texttt{head_positions} + 2n_L d_a p^2 \quad \text{and}$$
$$\texttt{flops_steps} = 2N_{\text{ne}}\,n + 2Vd\,\texttt{head_steps} + 2n_L d_a\left[(p+n)^2 - p^2\right];$$
$$\texttt{sequential_passes} = 1 + \texttt{latent_steps} + \texttt{emitted_tokens}.$$

The worked example (ProsQA Coconut, $p = 251$, 6 latents, 8 answer tokens) is 47,248,760,832 FLOPs and 15 passes, and the accounting test must reproduce it. The FLOP total charges 16 forward calls: the prefill of $p = |q| + 1$ positions, the 6 latent steps, the end-of-latent marker step, which produces the first answer token, and 8 emitted-token steps. Under the registered position table the marker step is charged in FLOPs but not counted as a sequential pass (it plays the role of the last prompt position), so a feedback arm's pass count is one lower than its forward calls; this favours the feedback arms, which fail the passes criterion regardless. E' is the mean number of tokens emitted before eos. Training FLOPs charge 3× forward per sequence per epoch. The GSM-50k pause and latent arms are also charged their CoT start, 47 charged epochs in all (the 25-epoch CoT run and 22 latent epochs), and every training ratio compares an arm with the bar's training FLOPs at the same charged epochs: on GSM-50k the bar's 64.8 PFLOPs over 25 epochs becomes $64.8 \times 47/25 = 121.8$ PFLOPs. An arm above 1.25× is "training-heavy". Wall-clock is measured at batch 1 on one named device and never compared across devices.

The head-cost confound is named in advance: a latent step costs $2N_{\text{ne}}$, a token $2N_{\text{ne}} + 2Vd$, so at 124M a latent step is 0.69 of a token pass (0.85 at 355M, 0.92 at 774M). At matched FLOPs a latent arm gets about 1.45 steps per CoT token at this size; this alone predicts an advantage that shrinks with scale.

3.6 Statistics

Paired differences use 10,000 paired example-bootstrap draws (numpy `default_rng(0)`, the same draw on every arm). The interval is $\Delta \pm 1.96\sqrt{\text{Var}_{\text{ex}}}$; at the screen both reads score one checkpoint, so no seed term enters. A full verdict (beats, loses, tie) needs $\sigma_{\text{task}} = \max(1.0,$ the tuned bar's seed SD) plus a determinism term from a same-seed pair. We measured it on ProsQA only: $\sigma_{\text{task}}(\text{ProsQA}) = 4.12$ (Section 4.2). No σ_{task} exists on any other task, so this paper contains no full verdict (beats, loses or tie) on any task; every Phase 2 result is a screen.

3.7 Governance

The protocol (tasks, arms, judge, kill rules, labels) was written and signed before any run. Amendments could only tighten a rule unless the project owner directed otherwise, and in that case the original verdict is still printed. An independent automated reviewer that wrote no model code registered every arm before its screen with numeric predictions (Appendix A) and recomputed every verdict from the hashed per-example files. Every result table in the repository was written once by a script from those files and reissued, never edited; every label in a table must appear verbatim in a reviewer verdict or the script refuses. The tables in this paper are copied from them, with the C1b FLOP ratios at the measured fraction added from the same records. Spend ran through a hash-chained ledger under a fixed cap, and every incident is logged (Appendix B). The protocol also forbids a set of words that overstate a diagnostic, and this paper keeps to it.

4 Reproduction (K1)

4.1 The oracle table

K1 passes on ProsQA (cot, no_cot, coconut, pause) and on GSM8k-Aug at one seed (cot 40.94, coconut 34.65 against 42.9 / 34.1, tolerances 2.0 / 3.0). ProntoQA, GSM8k-Aug `no_cot` and `coconut_no_curriculum` were not run in this phase, so K1 over ProntoQA is untested. On ProsQA our CoT cell (74.87, three seeds) scores below our `no_cot` cell (78.0, one seed), while the published cells sit the other way round (77.5 against 76.7); each of ours is within its own tolerance.

Three labels matter. The GSM `cot` cell is a single-seed match at the tolerance floor: 540 of 1,319, abs(diff) 1.96 of 2.0; 539 correct would have missed. The paper-style token count on `cot` is 30.6 against the published 25.0; the reference test chains themselves count 30.1 under our definition, so the published convention is unstated and the flag is a definition difference (Data: `data/table9_text_numbers.csv`). The ProsQA `cot` seed-1 audit (3 of 500 differ at batch 1) is explained as TF32 batch-shape rounding flipping near-ties along the greedy chain; a same-shape swap shows 0 of 300 cross-row differences and strict fp32 shows 0 of 300 divergences. The seed spread is our own measurement: the ProsQA `cot` SD over three seeds is 3.60 points, against the published ±1.9, and it later became a term of σ_{screen}.

4.2 Precision pair and determinism

$d_{\text{prec}} = 0.0$: ProsQA coconut seed 0 reads 96.8 at strict fp32 against 96.8 at fp32-tf32 (per-example agreement 476 of 500; Data: `data/table9_text_numbers.csv`), so the reproduction is fp32-equivalent on ProsQA; this is not claimed for GSM. The determinism term comes from two further runs of ProsQA `cot` at seed 0 with zero resumes (det1, det2), each read once on test with a batch-1 audit (Table 3).

Two readings of σ_{task}(ProsQA) are possible, with the ProsQA `cot` 3-seed SD 3.60 as the seed term and $0.2 \times 3.60 = 0.72$ as the line above which the determinism term counts. The strict reading, which we adopt, uses the zero-resume pair det1 / det2: $d_{\text{nd}} = |77.8 - 75.8| = 2.0$ (10 of 500 net), so $\sigma_{\text{task}} = \sqrt{3.60^2 + 2.0^2} = 4.12$. The alternative uses s0 / det1: $d_{\text{nd}} = 0.8$ (4 of 500 net), σ_{task} 3.69. The pair was read on two GPU types (det1 on A100-40GB, det2 on A100-80GB, both trained on A100-80GB), and det2's accuracy enters no cell, seed pool or σ_{cell}. The per-example churn between

Table 2: K1 against the published cells. Every cell: KNOWN (reproduction); fp32-tf32; ProsQA read on A100-40GB of A100-80GB checkpoints.

task	arm	seeds (test)	mean	SD	σ_{cell}	tolerance	published	verdict	labels
ProsQA	cot	75.0 / 71.2 / 78.4	74.87	3.60	2.88	5.76	77.50	matched	three seeds; audit label below
ProsQA	no_cot	78.0	78.00		1.00	2.00	76.70	matched	single seed
ProsQA	coconut	96.8 / 96.2 / 95.8	96.27	0.50	1.00	2.00	97.00	matched	three seeds
ProsQA	pause	97.0	97.00		1.00	2.00	96.60	matched	single seed
GSM8k-Aug	cot	40.9 (540 of 1,319)	40.94		1.00	2.00	42.90	matched	single seed; on the edge; secondary oracle flagged
GSM8k-Aug	coconut	34.6 (457 of 1,319)	34.65		1.50	3.00	34.10	matched	single seed

Data: `data/table1_k1_reproduction.csv`.

Table 3: The same-seed pair on ProsQA, test 500, each checkpoint read once. Labels: 1 seed per member (seed 0); pair read on two GPU types; det2's accuracy enters no cell, no seed pool and no σ_{cell}.

(a) Members.

member	run	resumes	selected epoch	test [95% interval] (count)	read on	trained on	batch-1 audit flips
s0	`prosqa_cot_s0`	1	46	75.0 [71.2, 78.8] (375 of 500)	A100-SXM4-40GB	A100-SXM4-80GB	0 of 500
det1	`prosqa_cot_det1_s0`	0	25	75.8 [72.0, 79.6] (379 of 500)	A100-SXM4-40GB	A100-SXM4-80GB	1 of 500
det2	`prosqa_cot_det2_s0`	0	23	77.8 [74.2, 81.4] (389 of 500)	A100-SXM4-80GB	A100-SXM4-80GB	0 of 500

(b) Per-example agreement (churn = examples that differ in correctness; net = the count difference).

pair	both correct	only the first	only the second	neither	churn	net
s0 / det1	324	51	55	70	106	−4
det1 / det2	341	38	48	73	86	−10
s0 / det2	333	42	56	69	98	−14

Data: `data/table1b_same_seed_pair.csv`, `data/table1b_same_seed_pair_agreement.csv`, `data/table1b_sigma_task_prosqa.csv`.

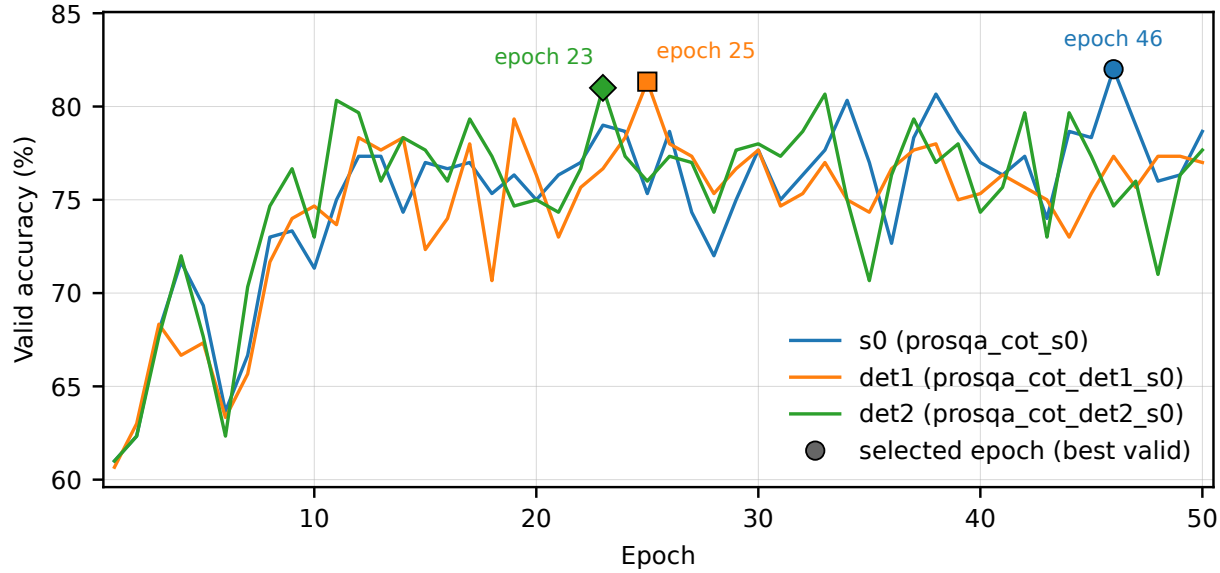


Figure 1: The same-seed pair on ProsQA: valid accuracy per epoch of `prosqa_cot_s0`, `prosqa_cot_det1_s0` and `prosqa_cot_det2_s0` over 50 epochs (valid 300); the marked points are the selected epochs (46, 25 and 23). The three valid curves are a diagnostic. Labels: 1 seed per member (seed 0). Data: `data/figure1_same_seed_pair_curves.csv`.

same-seed runs (86 to 106 of 500) is large relative to the net differences. The three valid curves are in Figure 1; they are a diagnostic, and d_{nd} is the test difference alone.

$\sigma_{\text{task}} = 4.12$ changes no table in this paper: the screens were read at $\sigma_{\text{screen}} = 3.60$, which carries no determinism term by design, and were not reissued. For transparency, four registered screen margins sit between 3.60 and 4.12 and would read differently at 4.12: on GSM-50k, C1b $k = 8$'s Δ_{scr} +4.0 [+1.70, +6.30] (K2 not fired at 3.60; fired at 4.12, and with it C1b $k = 8$'s pass line and screen scrambled leg would not be met); on ProsQA-hard, C1b $k = 4$'s pause comparison −4.0 (not within σ at 3.60; within at 4.12, so CURRICULUM-ONLY); and on GSM-50k the label "no better than plain depth" would attach to C3 (+4.02 over A_{pd}) and C3-foreign (+3.62). Every other registered outcome reads the same at either value.

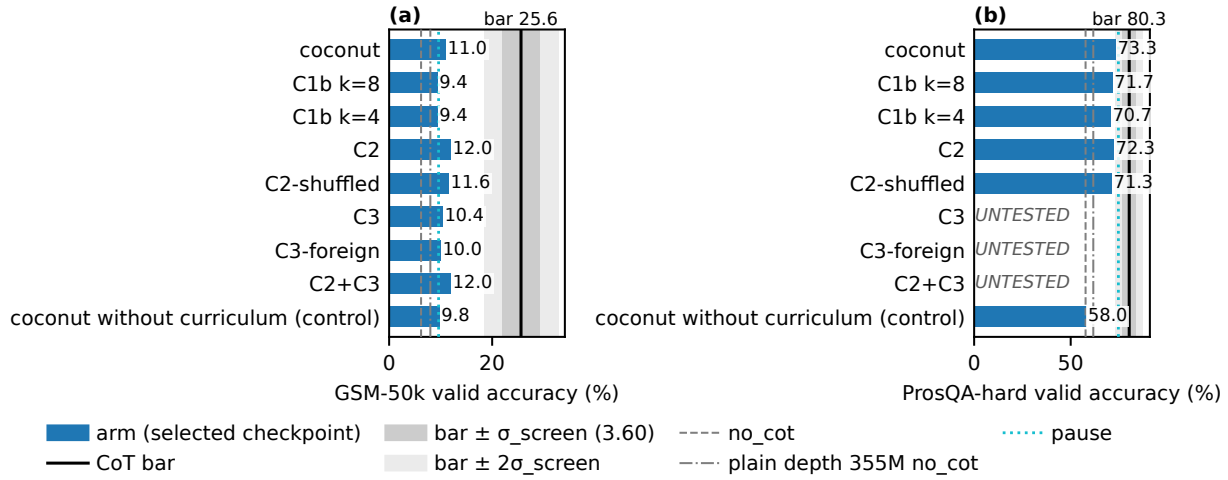


Figure 2: Both screens at seed 0: (a) GSM-50k, valid 500; (b) ProsQA-hard, valid 300. Each arm's selected checkpoint against the tuned bar (shaded: the bar ± σ_{screen} = 3.60 and ± 2 σ_{screen}), `no_cot`, plain depth 355M `no_cot` and pause. Labels: screen number (1 seed); valid only; no test read; bar LR-tuned 3 trials; arm 1 trial. A screen filters and is never a verdict. No card arm advances on either task. The C3 arms did not run on ProsQA-hard: untested. Data: `data/table2_gsm50k_screen.csv` and `data/table3_prosqa_hard_screen.csv`.

4.3 The Phase 1 diagnostic: idle on logic, load-bearing on arithmetic

ProsQA, three seeds, test 500. The three selected Coconut checkpoints' answers show no measurable dependence on their latents: scrambled drop 0.0 points on every seed (0 of 500 test examples change on each; paired 95% CI [0.0, 0.0]); zero-latent drop −0.4 / −0.2 / 0.0; step sweeps flat within 0.4 points from $n = 0$ to $n = 8$. The scrambled pass fed every example another example's latents, verified bit for bit on the valid split (no zero distance in 1,800 pairs per checkpoint). The latents are read by the model, since the next-token logits move on 300 of 300 valid examples (by up to 1.5 to 2.3 at the first emitted position), and they decide no emitted token on any of them. The latent vectors of different examples share a common direction (mean cosine 0.99 between an example's latent and its donor's at the same rank). A fourth checkpoint trained and read at strict fp32 shows the same. Data: `data/table7_step_sweep.csv` and `data/table9_text_numbers.csv`.

GSM8k-Aug, one seed, test 1,319. Coconut own 34.65; scrambled 3.94 (paired drop 30.71 [28.06, 33.35]); zero-latent 6.52 (28.13 [25.52, 30.73]); step sweep $n = 1$ 9.10, $n = 2$ 18.42, $n = 4$ 30.10, $n = 8$ 34.80 (out of distribution). The intervals are recomputed from the per-example files with the method of Section 3.6. Data: `data/table7_step_sweep.csv` and `data/table9_text_numbers.csv`.

The protocol registered the GSM numbers as printed, not interpreted, so the reading that follows is ours and is not a registered verdict: on the logic task the reproduced checkpoints answer the same with any example's latents, as recent critiques report; on the arithmetic task replacing the latents removes most of the accuracy. The same recipe at the same size produces latents that are idle under intervention on one task and load-bearing on the other. This agrees with Dilgren and Wiegreffe (2026) and is why the screens were run on both task families.

5 Screens

Every number in this section is a screen number: 1 seed (seed 0), valid only, no decision from test, bar LR-tuned over 3 trials, each arm 1 trial at lr 1e-4 (the disadvantaged side; Figure 2).

5.1 GSM-50k

Bar: `cot` lr 3e-4, 25.6 (128 of 500) at epoch 17 (trials at lr 3e-5 / 1e-4 / 3e-4: 21.0 / 23.8 / 25.6; Data: `data/table9_text_numbers.csv`). `no_cot` 6.2 (31 of 500). Plain depth 355M `no_cot` 8.0 (40 of 500), epoch 24, trained at micro-batch 16 × 8 after an out-of-memory probe at 32.

Screen rule. No arm advances: the bar leg fails on every card arm (9.4 / 9.4 / 12.0 / 10.4 / 12.0 against $25.6 - 3.60 = 22.00$), and so does the pause leg. The FLOP and token legs are met by every arm (E' 2.21 to 2.40 against $0.2 \times 30.44 = 6.09$). The passes criterion (≤ 6.29) is met by pause only, so every feedback arm is "billing claim only".

K2. Fired on C1b $k = 4$ (Δ_{scr} +1.4, interval including 0); not fired on coconut, C1b $k = 8$, C2 and C2-shuffled. C2's scrambled drop (+8.6) does not separate it from its null (+8.6) or from coconut (+7.0). The hinge arms are in Section 6. K2 also fired on the no-curriculum control (+1.6 [−0.06, +3.26]; Section 5.4).

K3. A_{pd} runs from 6.2 at [10.261, 10.269] GFLOPs (124M `no_cot`) to 8.0 at [35.948, 35.972] GFLOPs (355M); the latent arms sit at 11.074 to 11.689 GFLOPs (C1b at its measured fraction, as K3 requires), where A_{pd} is 6.31 to 6.39. K3 fires on no arm. Pause, C1b $k = 8$ and $k = 4$ are "no better than plain depth" (within 3.60 of A_{pd}), as is the no-curriculum control (+3.42 over A_{pd}); coconut, C2, C2-shuffled and C2+C3 sit 4.6 to 5.6 above it and C3 and C3-foreign +4.02 and +3.62, screen numbers against a 1-seed control.

Pause ties everything. Every latent arm is within 3.60 of the pause arm (differences −0.2 to +2.4): CURRICULUM-ONLY on all eight, and on the no-curriculum control (+0.2).

Earlier stage is better. Every latent arm's all-epoch best on valid sits at stage 1 (epochs 5 or 6, 2 latents): 11.0 to 14.6, above its final-regime selection (9.4 to 12.0). This is a flag, not a cell; the earlier-stage checkpoints emit 21 to 23 tokens and meet no token leg. The no-curriculum control has no earlier stage.

5.2 ProsQA-hard

Bar: `cot` lr 3e-5, 80.3 (241 of 300) at epoch 14 (trials at lr 3e-5 / 1e-4 / 3e-4: 80.3 / 70.7 / 73.0; Data: `data/table9_text_numbers.csv`); the bar sits at the LR-grid edge. `no_cot` 57.7 (173 of 300). Plain depth 355M `no_cot` 61.7 (185 of 300), epoch 28. The C3 arms did not run on this task (budget): untested.

Screen rule. No card arm advances (bar legs 71.7 / 70.7 / 72.3 against $80.3 - 3.60 = 76.73$; pause legs −3.0 / −4.0 / −2.3). FLOP and token legs are met by every arm (E' 8.15 to 8.18 against $0.2 \times 76.07 = 15.21$); the passes criterion ($\leq$ 15.41) is met by pause only.

K2. Fired on coconut, C1b $k = 4$, C2 and C2-shuffled: every interval includes 0, and the coconut drop is exactly +0.0 (0 of 300 examples change under another example's latents). Not fired on C1b $k = 8$ (+8.0 [+3.55, +12.45]), whose sweep rises from 64.0 at $n = 0$ to 72.3 at $n = 4$; the other four sweeps are flat within 1.3 points (Data: `data/table7_step_sweep.csv`).

Reading. Every curriculum-trained latent arm's selected checkpoint scores 13.0 to 15.7 points above `no_cot` (57.7). On four of the five feedback arms the scrambled read and the zero-latent read sit within 1.3 points of the own read, and pause, which feeds nothing back, is the highest arm; C1b $k = 8$ is the exception (scrambled −8.0, zero-latent −7.7, sweep rising to $n = 4$). So on those four arms the gain over `no_cot` does not depend measurably on what the fed-back vectors carry. The candidate the controls point to is the staged chain supervision that the pause arm shares, the curriculum's training data; the no-curriculum control (Section 5.4) tests it.

K3. A_{pd} is 57.69 to 57.72 (the 355M point costs 3.48× the 124M point and buys 4.0 points); every arm sits 12.97 to 16.95 above it (C1b at its measured fraction). K3 fires on nothing.

Earlier stage is better. Every latent arm's all-epoch best sits at an earlier stage (2 to 5 latents, 23 to 57 written tokens) above its final-regime selection by 3.3 to 9.3 points: coconut 76.7 at epoch 8, C1b $k = 8$ 75.0 at 14, $k = 4$ 74.3 at 18, C2 81.7 at 9, C2-shuffled 76.7 at 9. A flag, not a selection; none meets the token leg.

5.3 Arm verdicts across both screens

- **C1b $k = 4$:** killed at the screen (K2 fired on both tasks); CURRICULUM-ONLY on GSM-50k (pause −0.2), below pause by 4.0 on ProsQA-hard; f_{meas} nominal on both (0.4326 / 0.4391 against the line 0.43).
- **C2:** killed at the screen, within σ_{screen} of C2-shuffled on both tasks (+0.4 on GSM-50k, +1.0 on ProsQA-hard, against 3.60). Each gap includes the difference between a learnable and an unlearnable auxiliary target; the last supervised target contains the answer in 0.770 of the GSM-50k records and 1.00 of the ProsQA-hard records (Data: `data/table9_text_numbers.csv`). On neither task does the scrambled drop separate C2 from its null (+8.6 against +8.6; +0.3 against +0.3). We make no claim about what the decoder reads out: its positive control never ran.
- **C1b $k = 8$:** its own pass line is met on both tasks as a screen number: within σ_{screen} of full-depth coconut (−1.6 on GSM-50k, −1.7 on ProsQA-hard) with K2 not fired on either (+4.0 / +8.0), and the wall-clock line met on both (Section 7.3). The screen rule failed on both (bar legs −16.2 / −8.7; pause legs −0.2 / −3.0): it does not advance,

Table 4: GSM-50k screen, valid 500. Labels: screen number (1 seed); valid only; no test read; bar LR-tuned 3 trials; arm 1 trial.

arm	valid	vs bar	Δ_{scr} [interval]	K2 at the screen	zero-latent drop	pause comparison	passes; FLOP ratio to bar
pause	9.6	−16.0	(no fed-back vector; no K2 read)				3.210; [0.671, 0.673]
coconut	11.0	−14.6	+7.0 [+4.11, +9.89]	not fired	+10.0 [+7.24, +12.76]	+1.4: CURRICULUM-ONLY	9.396; [0.673, 0.675]; billing claim only
C1b $k = 8$	9.4	−16.2	+4.0 [+1.70, +6.30]	not fired	+1.4 [−0.90, +3.70]	−0.2: CURRICULUM-ONLY	9.280; [0.655, 0.657] at f_{meas} ([0.652, 0.654] nominal); billing claim only
C1b $k = 4$	9.4	−16.2	+1.4 [−0.12, +2.92]	fired	+1.2 [−0.03, +2.43]	−0.2: CURRICULUM-ONLY	9.284; [0.638, 0.640] at f_{meas} ([0.632, 0.634] nominal); billing claim only
C2	12.0	−13.6	+8.6 [+5.70, +11.50]	not fired	+11.0 [+8.19, +13.81]	+2.4: CURRICULUM-ONLY	9.310; [0.672, 0.674]; billing claim only
C2-shuffled	11.6	−14.0	+8.6 [+5.74, +11.46]	not fired	+10.0 [+7.00, +13.00]	+2.0: CURRICULUM-ONLY	9.230; [0.671, 0.673]; billing claim only
C3	10.4	−15.2	+10.4 [+7.71, +13.09]	not independent; counts as fired	+10.4 [+7.71, +13.09] (diagnostic)	+0.8: CURRICULUM-ONLY	9.226; [0.671, 0.673]; billing claim only
C3-foreign	10.0	−15.6	+2.0 [+0.00, +4.00]	not independent; fired on the drop leg	+2.6 [+0.49, +4.71] (diagnostic)	+0.4: CURRICULUM-ONLY	9.216; [0.671, 0.673]; billing claim only
C2+C3	12.0	−13.6	+11.8 [+8.99, +14.61]	not independent; counts as fired	+12.0 [+9.16, +14.84] (diagnostic)	+2.4: CURRICULUM-ONLY	9.268; [0.672, 0.673]; billing claim only
coconut without curriculum (control; section 5.4)	9.8	−15.8	+1.6 [−0.06, +3.26]	fired	+1.0 [−0.70, +2.70]	+0.2: CURRICULUM-ONLY	9.262; [0.672, 0.673]; billing claim only

Data: `data/table2_gsm50k_screen.csv`.

Table 5: ProsQA-hard screen, valid 300. Labels: screen number (1 seed); valid only; no test read; bar LR-tuned 3 trials; arm 1 trial.

arm	valid	vs bar	Δ_{scr} [interval]	K2 at the screen	zero-latent drop	pause comparison	passes; FLOP ratio to bar
pause	74.7	−5.7	(no fed-back vector; no K2 read)				9.153; [0.869, 0.872]
coconut	73.3	−7.0	+0.0 [+0.00, +0.00]	fired	+0.7 [−0.26, +1.59]	−1.3: CURRICULUM-ONLY	16.147; [0.869, 0.872]; billing claim only
C1b $k = 8$	71.7	−8.7	+8.0 [+3.55, +12.45]	not fired	+7.7 [+3.60, +11.73]	−3.0: CURRICULUM-ONLY	16.163; [0.866, 0.869] at f_{meas} ([0.865, 0.868] nominal); billing claim only
C1b $k = 4$	70.7	−9.7	−0.3 [−0.99, +0.32]	fired	+0.3 [−0.32, +0.99]	−4.0: not within σ	16.147; [0.863, 0.866] at f_{meas} ([0.862, 0.865] nominal); billing claim only
C2	72.3	−8.0	+0.3 [−0.32, +0.99]	fired	+0.7 [−0.26, +1.59]	−2.3: CURRICULUM-ONLY	16.180; [0.869, 0.872]; billing claim only
C2-shuffled	71.3	−9.0	+0.3 [−0.33, +1.00]	fired	+1.3 [+0.03, +2.64]	−3.3: CURRICULUM-ONLY	16.147; [0.869, 0.872]; billing claim only
coconut without curriculum (control)	58.0	−22.3	−1.0 [−2.74, +0.74]	fired	+1.0 [−0.47, +2.47]	−16.7: not within σ	16.240; [0.869, 0.872]; billing claim only

Data: `data/table3_prosqa_hard_screen.csv`.

is CURRICULUM-ONLY on both and billing claim only on both. We make no claim about what its latents carry. Its GSM-50k Δ_{scr} would not clear $\sigma_{\text{task}} = 4.12$ (Section 4.2).

- **C3 and C2+C3:** killed as latent claims on GSM-50k by K2 (counts as fired; Section 6); untested on ProsQA-hard.
- **Readout:** coverage and load-bearing fraction were measured on no arm (no probe, patch or bit-width ladder ran on any screen checkpoint). Every arm's readout is untested.

5.4 The no-curriculum control

Coconut without the curriculum, seed 0, under the same judge: its latent count is pinned at the final count at every trained epoch, and its selection runs over every trained epoch.

GSM-50k. The arm (6 latents at every epoch; selected at epoch 20; zero resumes) scores 9.8 (49 of 500): +3.6 over `no_cot` and −1.2 under full-depth coconut (11.0). It is within σ_{screen} of both, and the coconut-vs-`no_cot` gap on this task (4.8) is itself only 1.3 σ_{screen}, so the role of the curriculum on GSM-50k is unresolved at one seed. Its other reads: scrambled 8.2, zero-latent 8.8; step sweep $n = 0$ 8.8, $n = 1$ 9.0, $n = 2$ 9.8, $n = 4$ 9.2, $n = 6$ 9.8 (the trained count), $n = 8$ 9.8 (out of distribution); E' 2.262, 9.262 passes, 11.656 GFLOPs ([0.672, 0.673] of the bar): billing claim only. Its own K2 fired (Δ_{scr} +1.6 [−0.06, +3.26]; zero-latent drop +1.0 [−0.70, +2.70]), its pause comparison is +0.2 (CURRICULUM-ONLY), and K3 is not fired. Data: `data/table7_step_sweep.csv`.

ProsQA-hard. The arm (7 latents at every epoch; selected at epoch 2 of 30; one exact boundary resume) scores 58.0 (174 of 300): +0.3 over `no_cot` (57.7) and −15.3 under full-depth coconut (73.3). The same latent layout trained without the staged curriculum scores with `no_cot`, so at this read (1 seed) the curriculum is necessary for the latent arms' gain over `no_cot`. The coconut-vs-`no_cot` gap here (15.7) is 4.4 σ_{screen}. Its own K2 fired (Δ_{scr} −1.0 [−2.74, +0.74]; zero-latent drop +1.0 [−0.47, +2.47]), its pause comparison is −16.7 (not within σ_{screen}), and K3 is not fired (58.0 against A_{pd} 57.71: no better than plain depth). Data: `data/table7_step_sweep.csv`.

The two readings stay on their own tasks: on GSM-50k unresolved, on ProsQA-hard the curriculum is necessary, both at one seed. The published no-curriculum cells (ProsQA 76.1 ± 0.2, GSM8k-Aug 14.4 ± 0.8) are on the released tasks and enter no line here.

6 The dependence hinge (C3)

6.1 Design

C3 adds to Coconut a second forward pass per training batch in which every fed-back latent is replaced by a donor example's latent (a per-batch derangement, reseeded per epoch, latents detached). With CE_{true} and CE_{scr} the per-token answer cross-entropy under the two passes, the loss is $\text{CE}_{\text{true}} + \beta \max(0,\ m - (\text{CE}_{\text{scr}} - \text{CE}_{\text{true}}))$, $\beta = 1$, $m = 0.2$ nat per answer token. The hinge is satisfied only when the model's own latents make its answer at least m nats per token more likely than a stranger's latents do. It turns the scrambled control into a training signal and attacks the idle-latent failure mode directly. Its null, C3-foreign, computes the same hinge between two independent foreign banks, keeping the extra pass and the gradient into the latent path and removing the own-versus-foreign direction. Inference cost equals Coconut's; the training cost, against the bar at the same 47 charged epochs (Section 3.5), is 1.30× for C3, 1.68× for C3-foreign and 1.32× for C2+C3 (training-heavy).

The circularity was registered before the run: the hinge maximises exactly what K2 measures, and a model can satisfy it with a fingerprint. So the drop decides nothing by itself. The non-circular read on GSM-50k is the donor-answer read, designed before any number existed and registered by the reviewer with a numeric prediction.

6.2 The directional read

Under the registered scrambled pass (one Sattolo derangement of the 500 valid examples, seed 0, the donor's six latents fed bit for bit at every rank), the class is the 295 examples whose donor's last supervised step result equals the donor's answer; 6 pairs whose own answer equals the donor's are excluded; 289 are scored. R_{dir} is the rate at which the generated answer under donor latents equals the donor's answer; R_{own} is the same rate under own latents (a coincidence baseline, expected near 0.01); G_{chg} is the fraction of scored outputs that change to something that is neither the own prediction nor the donor's answer.

Table 6: The donor-answer read on GSM-50k, valid, 289 scored of 295. Labels: screen number, 1 seed; diagnostic entry; bf16 autocast over TF32 on; read and trained on A100-80GB; no test read; ProsQA-hard untested.

arm	Δ_{scr} [interval]	$R_{\text{dir}}(k)$	$R_{\text{own}}(k)$	Δ_{dir}	$G_{\text{chg}}(k)$ vs R_{dir}	status	K2 cell
C3	+10.4 [+7.71, +13.09]	0.000 (0)	0.014 (4)	−0.014 [−0.027, −0.000]	0.997 (288): exceeded	absent	not independent; counts as fired: drop without direction (fingerprint-consistent)
C3-foreign	+2.0 [+0.00, +4.00]	0.014 (4)	0.010 (3)	+0.003 [−0.003, +0.010]	0.481 (139): exceeded	absent	not independent; fired on the drop leg
C2+C3	+11.8 [+8.99, +14.61]	0.000 (0)	0.007 (2)	−0.007 [−0.016, +0.003]	0.993 (287): exceeded	absent	not independent; counts as fired: drop without direction (fingerprint-consistent)
coconut (reference, no hinge)	+7.0 [+4.11, +9.89]	0.024 (7)	0.007 (2)	+0.017 [−0.003, +0.038]	0.830 (240): exceeded	absent	diagnostic; decides nothing

Data: `data/table4_directional_read.csv`.

6.3 Detects but does not read

The hinge did its one job: under a stranger's latents C3 drops from 10.4 to 0.0 and C2+C3 from 12.0 to 0.2, while the null C3-foreign drops only from 10.0 to 8.0. The C3 minus C3-foreign drop difference is +8.4 [+5.11, +11.69], and C2+C3 minus C3-foreign +9.8 [+6.46, +13.14]: "dependence on own latents (content or fingerprint)" (Data: `data/table7_step_sweep.csv` and `data/table9_text_numbers.csv`).

But the dependence has no direction. Under the donor's latents the donor's answer appears 0 of 289 times on both hinge arms, below the coincidence baseline of 4 and 2; 288 and 287 of 289 outputs change to something that is neither the own answer nor the donor's. The non-hinge coconut reference shows 7 of 289, also absent. Every other pass condition of C3 is met on this task (drop above 10; accuracy within 3.60 of coconut; the drop exceeding C3-foreign's by more than 3.60), and the arm still fails because K2 counts as fired: a pass line written on the scrambled drop cannot be met by a drop that has no direction, since that drop is the quantity the hinge trained against.

What the read shows is a checkpoint that detects foreign latents and does not read their content. We call this a self-signature: the latents carry enough for the model to recognise its own, and the answer head declines under a mismatch. "Fingerprint-consistent" is a label, not a finding that the latents are a fingerprint; the read cannot tell a fingerprint from reasoning state the head declines to express under a mismatch. The zero-latent reads (C3 0.0; C2+C3 0.0; C3-foreign 7.4) are diagnostics that decide nothing for these arms. The accuracy leg carries no signal either: C3 −0.6, C2+C3 +1.0 and C3-foreign −1.0 against coconut's 11.0, all within 3.60; and all three are within 3.60 of pause (CURRICULUM-ONLY). C2+C3 adds +0.0 over C2 and +1.6 over C3.

The reviewer's predictions set before the read are in Appendix A; the directional outcome (P(at least one hinge arm "drop without direction") = 0.65) was on the right side, and the garbage fraction landed above every predicted interval on every arm, a recorded miss that changes no reading.

7 Cost

7.1 The three legs

The pattern is the same on both tasks. Every latent arm meets the FLOP leg (0.64 to 0.68 of the bar on GSM-50k, 0.86 to 0.87 on ProsQA-hard, C1b at its measured fraction) and the token leg (0.07 to 0.11 of the bar's emitted tokens). Every feedback arm fails the sequential-passes criterion: 9.216 to 9.396 passes against the line 6.29 on GSM-50k, 16.147 to 16.180 against 15.41 on ProsQA-hard. The pause arm, which feeds nothing back, meets it on both. The primary claim test itself was never reached, since no arm met the accuracy leg; the cost legs are printed so the reader can see what a win would have cost. A wall-clock comparison against the bar is untested: the bar has no timed read.

7.2 The token saving on losing arms

On GSM-50k the latent arms emit 2.2 to 2.4 tokens per answer against the bar's 30.44 on valid and 30.68 on test: a 13× token saving on arms that sit 11 to 14 points under the bar on test. On ProsQA-hard the bar emits 77.79 tokens on test against 8.2 for every latent arm. The saving is real and is the published shape of the result; at equal FLOPs it buys no accuracy here, and at equal sequential passes it does not exist for a feedback arm (9.4 passes against 3.2 for `no_cot`, which scores 6.2 against coconut's 11.0 on GSM-50k valid).

7.3 C1b: a measured per-step saving

The f_{meas} measurement compares the C1b checkpoint's latent pass against the full-depth coconut checkpoint run through the same band code at band [0, 12], batch 1, A100-40GB, torch 2.14.1+cu130 (Table 8).

For $k = 8$ the measured fraction is under the line on both tasks: a latent step through 8 of 12 blocks costs 0.7200 and 0.7280 of a full-depth latent pass in wall-clock at batch 1, that is 28.0% and 27.2% less, against the nominal 8/12 = 0.667. For $k = 4$ the fraction (0.4326 / 0.4391) sits just above the registered line 0.43, so its saving is "nominal". "Saving shows" means the wall-clock line is met at one seed on valid, on an arm that does not advance; it is not a statement about what any latent carries. The like-for-like check behind it: the band path at batch 32 agrees with the HF path on every example (0 of 500; 0 of 300); at batch 1 two examples per task flip, explained as bf16 near-tie flips across the padded shape; the first-token logits of the two paths agree to ≤ 2.3e-13 (GSM-50k) and ≤ 1.7e-12 (ProsQA-hard) at

Table 7: Cost per valid example (GFLOPs; latent arms exact means from the per-example files; the bar and `no_cot` keep exact ranges because they have no per-example read).

task	arm	emitted tokens E'	E' / bar	passes	passes / bar	FLOPs	FLOPs / bar	training FLOPs charged / bar at the same charged epochs
GSM-50k	cot bar	30.436	1.000	31.436	1.000	[17.312, 17.357]	[0.997, 1.003]	1.00
GSM-50k	no_cot	2.206	0.072	3.206	0.102	[10.261, 10.269]	[0.591, 0.593]	0.67
GSM-50k	plain depth 355M	2.332	0.077	3.332	0.106	[35.948, 35.972]	[2.071, 2.078]	1.91 (training-heavy)
GSM-50k	pause	2.210	0.073	3.210	0.102	11.643	[0.671, 0.673]	0.92
GSM-50k	coconut	2.396	0.079	9.396	0.299	11.689	[0.673, 0.675]	0.92
GSM-50k	C1b $k = 8$	2.280	0.075	9.280	0.295	11.315 nominal; 11.371 at f_{meas}	[0.655, 0.657] at f_{meas}; [0.652, 0.654] nominal	0.91
GSM-50k	C1b $k = 4$	2.284	0.075	9.284	0.295	10.972 nominal; 11.074 at f_{meas}	[0.638, 0.640] at f_{meas}; [0.632, 0.634] nominal	0.90
GSM-50k	C2 / C2-shuffled	2.310 / 2.230	0.076 / 0.073	9.310 / 9.230	0.296 / 0.294	11.668 / 11.648	[0.672, 0.674] / [0.671, 0.673]	0.93 / 0.93
GSM-50k	C3 / C3-foreign / C2+C3	2.226 / 2.216 / 2.268	0.073 / 0.073 / 0.075	9.226 / 9.216 / 9.268	0.293 / 0.293 / 0.295	11.647 / 11.644 / 11.657	[0.671, 0.673] each (C2+C3 [0.672, 0.673])	1.30 / 1.68 / 1.32 (training-heavy)
ProsQA-hard	cot bar	76.070	1.000	77.070	1.000	[128.523, 128.960]	[0.997, 1.003]	1.00
ProsQA-hard	no_cot	8.190	0.108	9.190	0.119	[110.239, 110.376]	[0.855, 0.859]	0.90
ProsQA-hard	plain depth 355M	8.180	0.108	9.180	0.119	[384.111, 384.477]	[2.979, 2.992]	2.57 (training-heavy)
ProsQA-hard	pause	8.153	0.107	9.153	0.119	112.014	[0.869, 0.872]	0.95
ProsQA-hard	coconut	8.147	0.107	16.147	0.210	112.012	[0.869, 0.872]	0.95
ProsQA-hard	C1b $k = 8$	8.163	0.107	16.163	0.210	111.568 nominal; 111.651 at f_{meas}	[0.866, 0.869] at f_{meas}; [0.865, 0.868] nominal	0.94
ProsQA-hard	C1b $k = 4$	8.147	0.107	16.147	0.210	111.115 nominal; 111.258 at f_{meas}	[0.863, 0.866] at f_{meas}; [0.862, 0.865] nominal	0.94
ProsQA-hard	C2 / C2-shuffled	8.180 / 8.147	0.108 / 0.107	16.180 / 16.147	0.210 / 0.210	112.021 / 112.012	[0.869, 0.872]	0.95 / 0.95

Data: `data/table5_cost_per_example.csv`.

float64, with the strict-fp32 GPU figures 6.4e-4 and 7.6e-4 printed beside them; the registered line for this check was a largest logit difference of 1e-4, which the strict-fp32 figures miss.

Per example at batch 1 (timed passes only), C1b $k = 8$ takes 83.82 ms on GSM-50k against 103.77 ms for the full-depth denominator read, and 156.73 ms against 176.44 ms on ProsQA-hard. These compare to no bar.

7.4 Pareto read

The Pareto read is a compilation from the recorded per-example files, with no new training and no new read of any checkpoint. It places 135 points (GSM-50k 61 valid and 21 test; ProsQA-hard 39 valid and 14 test) on accuracy against mean emitted tokens E', mean FLOPs and mean sequential passes, valid and test kept apart (Figures 3 and 4; Tables 9 and 10). It is secondary and decides nothing.

Every gap in the Pareto read is descriptive (1 seed; test points from the single end-of-round test read). A FLOP multiplier and a budgeted-CoT curve, which would place written CoT at these token counts, are untested.

8 Test read (reporting only)

The screen checkpoints were read once on test at the end of the round, with no decision attached. The interval is ±1.96 binomial SE for one accuracy; test minus valid is descriptive, since valid is the selection split and test the scoring split.

Secondary reads (the all-epoch best checkpoint at its own stage's latent count; decides nothing): on GSM-50k the nine stage-1 checkpoints read 11.8 to 13.9 on test with 21 to 23 emitted tokens; on ProsQA-hard the five earlier-stage checkpoints read 71.2 to 78.8 with 25 to 57 emitted tokens. Of the four largest test minus valid gaps (the GSM-50k bar −4.6, the ProsQA-hard `no_cot` −4.5, C2 +3.5 and the ProsQA-hard bar +3.1), three are inside the full width of their test interval; the GSM-50k bar's is 0.2 outside (width 4.4). The bar's test drop narrows its lead on GSM-50k but every latent arm still sits 10.8 to 13.5 points under it on test. None is a result, and no line of the screens changes with it. Test emitted tokens: the GSM-50k bar 30.68 against 2.21 to 2.35 for the latent arms; the ProsQA-hard bar 77.79 against 8.17 to 8.20.

9 Limitations

1. **One seed.** Every Phase 2 number is seed 0; the bars are single-seed and LR-tuned only (the schedule axis untested); no confirmation seed ran. Four of the six K1 cells are single seed, and the GSM `cot` cell sits at the tolerance floor (540 of 1,319; 539 would have missed).
2. **Valid only.** Screen decisions use the selection split; the test read is reporting only. A screen filters and never gives a verdict: no beats / loses / tie verdict exists on any task.
3. **124M only.** No slope, no FLOP multiplier, no size above 124M except the 355M `no_cot` control (1 seed, micro-batch 16×8). The head-cost confound favours latent steps at this size.
4. **The bars.** LR-tuned over 3 trials against 1 trial for each arm; the ProsQA-hard bar sits at the bottom edge of the LR grid (3e-5) and the GSM-50k bar at the top edge (3e-4), and the grid was not extended beyond either, so both bars may be under-tuned, which only strengthens the negative; the released-ProsQA tuned bar was never read; selection on seed 0 inflates every bar slightly, against the latent arms.

Table 8: The f_{meas} measurement: the C1b checkpoint's latent pass against the full-depth coconut checkpoint run through the same band code at band [0, 12], batch 1, A100-40GB, torch 2.14.1+cu130.

task	arm	latent ms per pass	denominator (coconut, band [0, 12])	f_{meas} at batch 1	registered line	at batch 32	label
GSM-50k	C1b $k = 8$	8.006	11.120	0.7200	0.7667	0.7396	saving shows
GSM-50k	C1b $k = 4$	4.810	11.120	0.4326	0.43	0.4785	nominal
ProsQA-hard	C1b $k = 8$	8.053	11.063	0.7280	0.7667	0.7499	saving shows
ProsQA-hard	C1b $k = 4$	4.858	11.063	0.4391	0.43	0.5291	nominal

Data: `data/table8_c1b_wallclock.csv`.

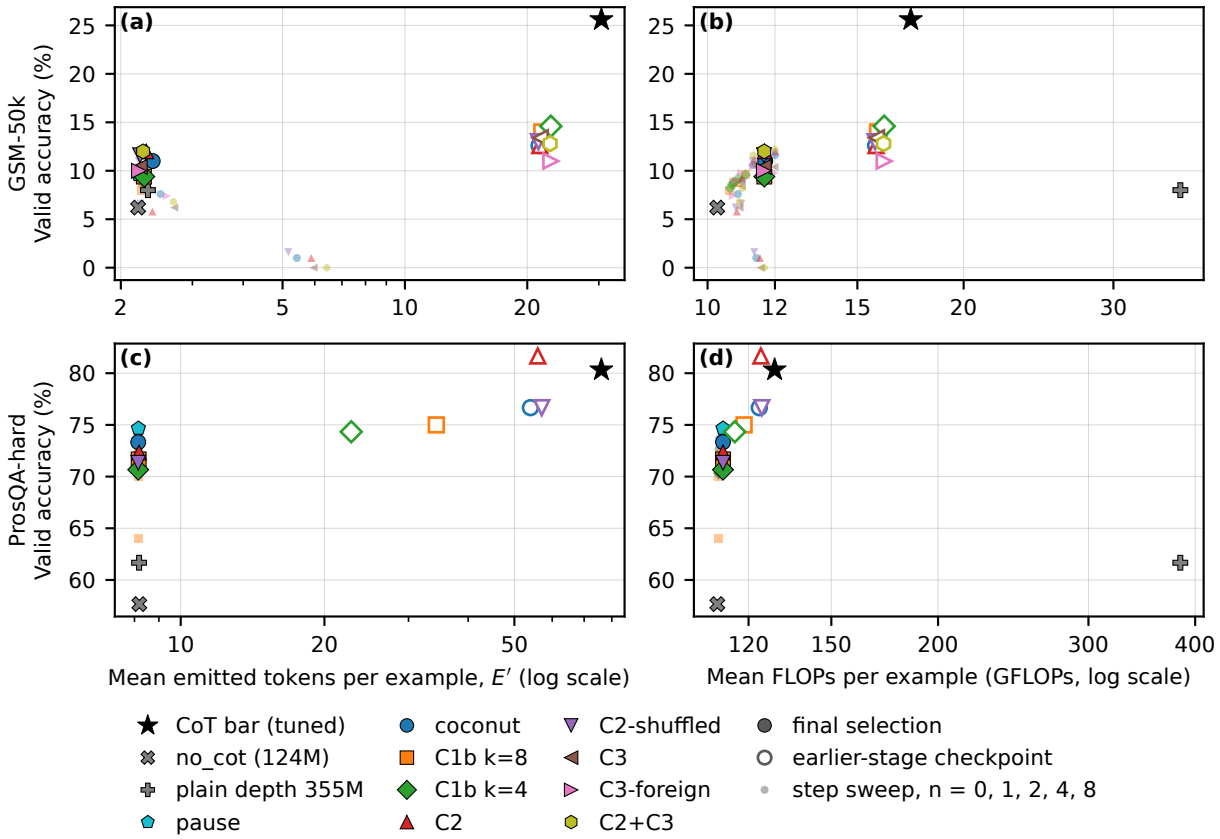


Figure 3: The Pareto read on valid: accuracy against mean emitted tokens E' (a, c) and against mean FLOPs (b, d), GSM-50k (a, b) and ProsQA-hard (c, d), cost axes on a log scale. Labels: 1 seed; decides nothing; the earlier-stage points and the bar come from the training log's means (range), with no per-example record on valid. On valid every earlier-stage point (14 of 14) sits above its final selection on accuracy, to its right on tokens and above it on FLOPs. Every gap is descriptive. Data: `data/figure2a_pareto_points_valid.csv`.

Table 9: The Pareto read's ranges on valid (accuracy in points; E' in tokens; mean GFLOPs). Labels: 1 seed; decides nothing; the earlier-stage points and the bar come from the training log's means (range), with no per-example record on valid.

task	points	accuracy	E'	GFLOPs
GSM-50k	nine earlier-stage checkpoints (stage 1, 2 latents, epochs 5 to 6)	11.0 to 14.6	21.3 to 22.9	15.7 to 16.2
GSM-50k	their final selections	9.4 to 12.0	2.2 to 2.4	11.6 to 11.7
GSM-50k	tuned bar	25.6	30.4	17.3
ProsQA-hard	five earlier-stage checkpoints (2 to 5 latents, epochs 8 to 18)	74.3 to 81.7	22.8 to 57.1	115.5 to 124.6
ProsQA-hard	their final selections	70.7 to 73.3	8.1 to 8.2	112.0
ProsQA-hard	tuned bar	80.3	76.1	128.5 to 129.0

Data: `data/figure2a_pareto_points_valid.csv`.

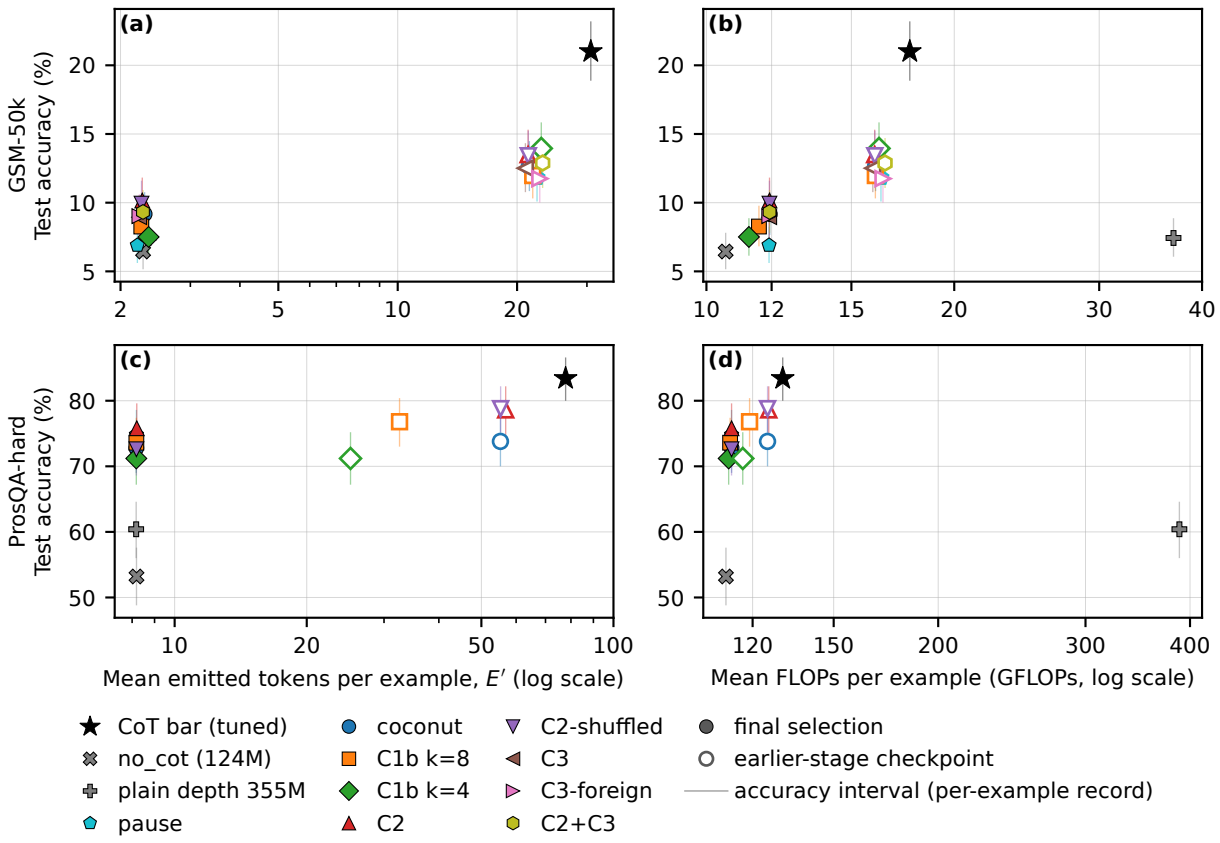


Figure 4: The Pareto read on test: accuracy against mean emitted tokens E' (a, c) and against mean FLOPs (b, d), GSM-50k (a, b) and ProsQA-hard (c, d), cost axes on a log scale; each point with a per-example record carries its accuracy interval. Labels: 1 seed; decides nothing; end-of-round test read: reporting only, no decision attached. On test every earlier-stage point sits to the right of its final selection on tokens and above it on FLOPs on 14 of 14, and above it on accuracy on 13 of 14 (C1b $k = 4$ on ProsQA-hard: both test points 71.2). Every gap is descriptive. Data: `data/figure2b_pareto_points_test.csv`.

Table 10: The Pareto read's ranges on test (accuracy in points; E' in tokens; mean GFLOPs). Labels: 1 seed; decides nothing; end-of-round test read: reporting only, no decision attached.

task	points	accuracy	E'	GFLOPs
GSM-50k	nine earlier-stage checkpoints	11.8 to 13.9	21.0 to 23.2	15.9 to 16.5
GSM-50k	their final selections	6.9 to 10.2	2.2 to 2.4	11.3 to 11.9
GSM-50k	tuned bar	21.0	30.7	17.7
ProsQA-hard	five earlier-stage checkpoints	71.2 to 78.8	25.2 to 56.8	116.8 to 125.5
ProsQA-hard	their final selections	71.2 to 75.8	8.2	112.3 to 113.2
ProsQA-hard	tuned bar	83.4	77.8	130.4

Data: `data/figure2b_pareto_points_test.csv`.

Table 11: End-of-round test read: reporting only, no decision attached. Primary reads: the selected `ckpt_best_final.pt` at the final stage's latent count.

run	GSM-50k test (1,319)	valid	ProsQA-hard test (500)	valid
tuned bar (cot)	21.0 [18.8, 23.2]	25.6	83.4 [80.1, 86.7]	80.3
no_cot	6.4 [5.1, 7.8]	6.2	53.2 [48.8, 57.6]	57.7
plain depth 355M	7.4 [6.0, 8.8]	8.0	60.4 [56.1, 64.7]	61.7
pause	6.9 [5.5, 8.3]	9.6	74.8 [71.0, 78.6]	74.7
coconut	9.2 [7.6, 10.7]	11.0	72.8 [68.9, 76.7]	73.3
C1b $k = 8$	8.3 [6.8, 9.7]	9.4	73.6 [69.7, 77.5]	71.7
C1b $k = 4$	7.5 [6.1, 8.9]	9.4	71.2 [67.2, 75.2]	70.7
C2	10.2 [8.5, 11.8]	12.0	75.8 [72.0, 79.6]	72.3
C2-shuffled	9.9 [8.3, 11.5]	11.6	72.6 [68.7, 76.5]	71.3
C3	8.9 [7.4, 10.5]	10.4	untested	
C3-foreign	9.0 [7.5, 10.6]	10.0	untested	
C2+C3	9.3 [7.8, 10.9]	12.0	untested	

Data: `data/table6_test_read.csv`.

5. **σ_{task} on ProsQA only.** σ_{task}(ProsQA) = 4.12 (3.69 under the alternative reading); no σ_{task} exists on any other task. The screens were read at σ_{screen} 3.60, and three registered margins would read differently at 4.12 (Section 4.2). The determinism pair was read on two GPU types.
6. **The readout is untested.** No probe, counterfactual patch or bit-width ladder ran on any screen checkpoint; the positive control never ran; "bits used per step" was never measured.
7. **The C2 target confound.** The last supervised state target contains the answer in 0.770 of GSM-50k and 1.00 of ProsQA-hard records; attribution rests on C2-shuffled, whose gap includes learnable against unlearnable targets.
8. **The hinge arms ran on one task.** C3, C3-foreign and C2+C3 are untested on ProsQA-hard.
9. **Numerics.** Phase 2 ran bf16 autocast over TF32; two examples per task flip between batch 1 and batch 32, explained at float64; the strict-fp32 band-logits records missed the registered 1e-4 line (largest logit difference between the band path and the reference path; Section 7.3) and are printed beside their float64 explanations.
10. **Untested, not estimated:** ProntoQA; the no-curriculum control beyond one seed; further cards; larger scales; program-output prediction; a long fade run; GSM8k-Aug `no_cot`; the 355M control's other seeds; wall-clock against the bar; the FLOP multiplier and budgeted-CoT curve.

10 Discussion

What the negative says. Within its scope the result is clean. The reproduction holds, so the methods were run as published. On two harder tasks no latent arm, with or without a new objective, comes within the screen's σ of a tuned written-CoT model at 0.64 to 0.87 of its FLOPs; every one of them ties or trails a pause-token model trained on the same curriculum. Our prior put this outcome at 0.60.

Dependence can be trained in; content cannot, with these methods. The hinge works as an objective: the model's answers collapse when its latents are swapped, while the content-free null's do not. But the directional read shows the model never takes anything from a stranger's latents; it only notices they are not its own. A self-signature is the cheapest way to satisfy a hinge on $\text{CE}_{\text{scr}} - \text{CE}_{\text{true}}$, and that is what gradient descent found. Verifiable-state supervision (C2) did not change this: it adds +0.4 and +1.0 over its shuffled null, and its scrambled drop equals the null's on both tasks. Neither objective moved accuracy against the bar.

The curriculum as the candidate cause. On ProsQA-hard the three full-depth feedback arms (coconut, C2, C2-shuffled) and C1b $k = 4$ answer the same with any example's latents, with no latents, and with eight; the pause arm, which feeds nothing back, is the highest arm; every curriculum-trained latent arm sits 13.0 to 15.7 points above `no_cot`. The controls point at the staged chain supervision that every latent arm and the pause arm share. The direct evidence is our no-curriculum control (Section 5.4): the same latent layout trained without the staged curriculum scores with `no_cot` (58.0 against 57.7, 15.3 below coconut), so at one seed the curriculum is necessary for the latent arms' gain on ProsQA-hard. An unreviewed preprint reports the same on released ProsQA (Anonymous, 2026); we see it at depth 5 to 7 with 1.7× the edges. On GSM-50k the same control is unresolved at one seed. On GSM-50k the latents are

load-bearing on coconut, C2, C2-shuffled and the two hinge arms (scrambled drops +7.0 to +11.8; C3-foreign +2.0; C1b +4.0 and +1.4) and every arm still ties pause within 3.60: carrying state and using it well are different things.

Earlier is better. On both tasks every latent arm's best valid accuracy sits at an earlier curriculum stage, with 2 to 5 latents and 21 to 57 written tokens, above its final-regime selection by up to 9.3 points. Each stage that replaces written steps with latents costs accuracy at this size and saves tokens and FLOPs. In the Pareto read, on valid, every earlier-stage point (14 of 14) sits above its final selection on accuracy, to its right on tokens and above it on FLOPs. On test the same holds for tokens and FLOPs on 14 of 14 and for accuracy on 13 of 14 (C1b $k = 4$ on ProsQA-hard: both test points 71.2). Every gap is descriptive; the budgeted-CoT curve is untested.

What a positive would need. Under this judge a positive read at 124M would need an arm within σ of a 3-LR bar on valid, ahead of pause and of its scrambled null by σ, with the token leg held, then three fresh seeds, a directional read that is present, a readout with coverage and load-bearing fraction above 0.5, and no fade over a long run. None of the first conditions was met by any arm. The methods that close the gap in the literature (CODI, SIM-CoT, LOTUS) supervise each latent against the explicit step or widen the per-step state; C2 did the former with executed state and did not move. Our prediction written before any run was that single-vector Coconut on GPT-2 would not beat tuned CoT on any task, and that a pass would need wider or cheaper steps with per-step supervision and a dependence loss. The second half was tested and did not pass either.

The one cost result. C1b $k = 8$ is the one positive line a later design can build on: a latent step through the middle 8 of 12 blocks keeps full-depth accuracy within σ on both tasks and costs 0.7200 to 0.7280 of a full-depth latent pass at batch 1. Layer skipping is itself a known idea (Fan et al., 2020; Elhoushi et al., 2024); what is measured here is that it holds for latent steps of this kind. It is a saving per step on an arm that does not advance, and the saving is in wall-clock and FLOPs, not in sequential passes.

A proposed next objective. Supervise content and penalise the signature. The hinge rewards any difference between own and foreign latents; the next objective should reward only the difference that carries the donor's content. Concretely: train against a donor whose intermediate state is known (the C2 target), and ask that under the donor's latents the model's answer move toward the donor's answer, not merely away from its own. The directional read of Section 6 becomes the training signal, and the donor-answer rate becomes the pass line. The idea is close to FRODO's counterfactual loss for written rationales (Paul et al., 2024) and to interchange intervention training (Geiger et al., 2022), here applied to fed-back latents. This is a proposal; it was not run.

Code and data

Code, the protocol, the per-example result files and the CSV behind every table and figure (named in each caption) are available at https://github.com/atilavahedian/silent-reasoning. No checkpoint is released.

Use of AI assistance

AI tools assisted with designing and writing the code, running the experiments and drafting the text. The author directed the study and is responsible for all content.

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A Registered predictions against outcomes

The predictions of the independent automated reviewer that wrote no model code, made before each run, and the outcomes.

Table 12: The registered predictions against outcomes.

prediction	outcome
ProsQA coconut 95.5 [92, 98]	96.27
ProsQA cot 78 (SD 2.0)	74.87
P(K1 ProsQA) 0.70	pass

Table 12, continued

prediction	outcome
GSM8k-Aug cot 42	40.94
GSM8k-Aug coconut 32 [27, 36]	34.65
P(K1 GSM) 0.50	pass (one seed)
ProsQA scrambled drop 4 [1, 20]; zero-latent 8 [2, 25]	0.0 on every seed; −0.4 / −0.2 / 0.0 (below the intervals)
ProsQA-hard bar 60 [40, 80]	80.3, above the interval
ProsQA-hard no_cot 55	57.7
ProsQA-hard coconut 70 [50, 90]	73.3
ProsQA-hard pause within 3 of coconut	+1.3, yes
GSM-50k bar 30 [24, 36]	25.6
GSM-50k no_cot 12	6.2
GSM-50k coconut from cot 22 [15, 30]	11.0, below
P(K2 fires on full-depth coconut on GSM-50k) 0.4	did not (it fired on ProsQA-hard)
C1b $k = 8$ minus coconut: −1 [−4, +2] (ProsQA-hard) / −2 [−5, +1] (GSM-50k)	−1.7 / −1.6
C1b $k = 4$ minus coconut: −5 [−10, 0] / −6 [−12, −1]	−2.7 / −1.6
P(K2 fires on $k = 8$) 0.4	did not on either task
wall-clock within 10% of 8/12: P 0.5	0.7200 / 0.7280, yes
C2 minus coconut 0 [−3, +2] / +2 [−2, +6]	−1.0 / +1.0
C2 minus C2-shuffled +1 [−2, +4] / +2 [−1, +5]	+1.0 / +0.4
C2's scrambled drop on GSM-50k 8 [2, 20]	8.6
P(C2 passes) 0.45	killed
C3 on GSM-50k: drop above 15 (P 0.45)	10.4, not met
C3 accuracy vs coconut +1 [−4, +4]	−0.6, inside
P(C3-foreign's drop exceeds 10) 0.25	2.0, not met
C2+C3 minus coconut +3 [−2, +9]	+1.0, inside
P(C2+C3 dead by K2) 0.2	dead (K2 counts as fired)
R_{own} 0.010 [0.003, 0.025] on every arm	met
Δ_{dir} C3 +0.03 [−0.01, +0.12] (P(present) 0.35)	−0.014, outside on the low side, absent
Δ_{dir} C2+C3 +0.05 [−0.01, +0.20]	−0.007, inside, absent
Δ_{dir} C3-foreign +0.01 [−0.01, +0.05]; coconut +0.02 [−0.01, +0.08]	+0.003, absent; +0.017, absent
G_{chg} C3 0.30 [0.10, 0.70]; C2+C3 0.25 [0.08, 0.60]; C3-foreign 0.12 [0.03, 0.40]; coconut 0.10 [0.03, 0.30]	0.997, 0.993, 0.481, 0.830, outside every interval on the high side (the reviewer's miss, recorded; it changes no reading)
P(at least one hinge arm "drop without direction") 0.65	met (two)
P(a card passes its screen on at least one task) 0.6	none advanced
P(the 124M read is positive under the primary claim's definition) 0.15	it is not

Prior before any run: P(full goal) about 0.10; P(tie with tuned CoT at equal FLOPs with 3–5× fewer emitted tokens) about 0.30; P(clean negative) about 0.60.

B Incidents and compute

Compute. All runs used spot A100 GPUs (40GB and 80GB); total compute was a few hundred dollars under a fixed cap.

Incidents. Spot preemptions hit every run group, including two storms on one day (seven preemptions and two stockouts in one hour). One bar trial (`prosqa_cot_lr3e-5_s0`) reached 5 resumes and is labelled "fragmented" (reported, not void); the ProsQA-hard screen spent its automatic relaunch counter (11 attempts) and was relaunched by hand under the same admission; every resume was from the registered checkpoint and no superseded attempt enters any number. The plain-depth 355M control ran at micro-batch 16×8 after its probe ran out of memory at 32×1024. The like-for-like check of the C1b band code differed on 2 of 500 (GSM-50k) and 2 of 300 (ProsQA-hard) examples at batch 1, explained as bf16 near-tie flips across the padded shape and confirmed at float64; the strict-fp32 band-logits records (6.4e-4, 7.6e-4) missed the 1e-4 line and stay printed beside their float64 explanations (2.3e-13, 1.7e-12). No checkpoint, read or number was lost. Data: `data/table9_text_numbers.csv`.

Known biases, recorded before or during the runs: selection of the bar on seed 0 inflates it slightly (against the latent arms); the GSM valid set shares 13-grams with train on 7 of 500 items (equal for every arm); the head-cost confound favours latent steps at small sizes; if a published value were a single run, our three-seed mean would differ from it with SD about $\sqrt{1 + 1/3}\,\sigma \approx 1.15\sigma$ under a perfect reproduction, so a 2 σ K1 tolerance would miss about 8% of cells ($P(|Z| > 2/1.155) = 0.083$); against a three-run published mean the rate is about 1.4%.

C Labels

- **screen number (1 seed); valid only; no test read; bar LR-tuned 3 trials; arm 1 trial:** every Phase 2 number. A screen filters and is never a verdict.
- **KNOWN (reproduction):** every Phase 1 cell; the reproduction is the oracle, not a contribution.
- **fp32-tf32:** read with TF32 matmuls on, the checkpoint's own numerics; the reference's precision is probably strict fp32; $d_{\text{prec}} = 0.0$ on ProsQA.
- **single seed / three seeds / 1 seed:** the seed count behind the number; two agreeing seeds are not a replication.
- **untested:** not run; never estimated. Distinct from a miss.
- **CURRICULUM-ONLY:** the arm is within σ of the pause-same-curriculum arm; its gain is attributed to the curriculum, not to the fed-back content.
- **K2 fired at the screen:** Δ_{scr} under σ_{screen} or its interval including 0; the arm's latent claim is killed on that task; it continues only as a labelled curriculum or token-budget effect.
- **K2 not independent:** the arm was trained against scrambled latents; its drop is not evidence of state.
- **K2 counts as fired:** a large drop whose directional read is absent or whose garbage fraction exceeds its directional fraction; labelled "drop without direction (fingerprint-consistent)".
- **fired on the drop leg:** K2 fired on $\Delta_{\text{scr}} < \sigma_{\text{screen}}$ alone (C3-foreign).
- **dependence on own latents (content or fingerprint):** the C3 minus C3-foreign drop difference.
- **no better than plain depth:** within σ of A_{pd} at the arm's FLOPs; K3 not fired.
- **billing claim only:** the token leg met and the sequential-passes criterion not met.
- **nominal:** a C1b arm whose measured per-step fraction is above the wall-clock line; K3 and the primary test use the measured fraction.
- **saving shows:** the wall-clock line met on the card's wall-clock number, one seed, valid, on an arm that does not advance; not a K3 outcome, not a primary-test claim, not a statement about what any latent carries, not a test number.
- **training-heavy:** training FLOPs above 1.25× the bar's at the same epochs.
- **earlier stage better on valid:** the arm's all-epoch best sits at an earlier curriculum stage; a flag, not a cell.
- **on the edge:** a K1 cell within one example of its tolerance (the GSM `cot` cell).
- **diagnostic; decides nothing:** a read registered as a diagnostic (step sweeps, zero-latent reads of hinge arms, the coconut reference entry of the donor-answer read).
- **end-of-round test read: reporting only, no decision attached:** the single test read of every screen checkpoint.
- **fragmented:** a run with more than a registered number of resumes; reported, not void.
- **pair read on two GPU types:** the same-seed pair's read leg (det1 on A100-40GB, det2 on A100-80GB, both trained on A100-80GB); the type printed per member.
- **strict / alternative:** the two readings of σ_{task}(ProsQA): strict on the pair det1 / det2 (4.12), the alternative reading on s0 / det1 (3.69).
- **from the training log's means (range); no per-example record on valid:** a Pareto point whose costs come from the training log, not from a per-example read; it carries no band.